Total No. of pages:1 | *Roll No.................*

Degree: B.Tech. Semester: III - Course work

MID-SEMESTER EXAMINATION, September 2024

Course Title: *Mathematics for Communication and Signal Processing*

Course Code: ECMTC301

Time: 1.5 Hours | Max. Marks: 20

Note: - Attempt all questions in given order only. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1			
1a	Show that $S = \{3^n : n \in \mathbb{Z}\}$ is a commutative group with respect to multiplication.	2	CO1
1b	Prove that a group G is abelian iff $(ab)^{-1} = a^{-1}b^{-1}$.	2	CO1
Q 2			
2a	Show that under homomorphism the direct image of a subgroup is a subgroup.	2	CO1
2b	Define cyclic group. Show that the homomorphic image of a cyclic group is a cyclic group.	2	CO1
Q 3			
3a	Define Ring. Show that $(\mathbb{Z}, +, .)$ is a ring.	2	CO1
3b	Define Field. Prove a $(\mathbb{Z}_5, +, .)$ is a field.	2	CO1
Q 4			
4a	Define subspace of a vector space. Is W a subspace of $\mathbb{R}^3$? If $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid 5a_1^2 - 3a_2^2 + 6a_3^2 = 0\}$	2	CO2.
4b	Define linear independent and linear dependent set of vectors. Is a set of vectors (x_1, x_2, x_3, x_4) in $\mathbb{R}^4$, where the vectors $x_1 = (1, 0, 1, 2)$, $x_2 = (0, 1, 1, 2)$, $x_1 = (1, 1, 1, 3)$ linear independent?	2	CO2
Q 5			
5a	Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(a_1, a_2) = (2a_1 + a_2, a_1)$. Show that T is linear transformation.	2	CO2
5b	If $T : V \longrightarrow W$ is a linear transformation then show that $dim\ V = Rank\ T + Null\ T$.	2	CO2